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Design-to-fabrication workflow for 3D concrete printed structures with embedded periodic anticlastic surfaces

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ABSTRACT

This paper presents an integrated computational design and digital fabrication workflow, creating tension and compression combined spanning structures through 3D Concrete Printing (3DCP). The approach combines Polyhedral Graphic Statics (PGS) for structural form-finding, and the optimized periodic anticlastic geometries inspired by Triply Periodic Minimal Surfaces (TPMS) for material distribution. The PGS implementation enables visualization and control of force flow in three dimensions, allowing precise allocation of materials—depositing concrete in compression zones and steel cables in tension regions. 3DCP's technical constraints are addressed by developing algorithms that minimize local overhang angles and generate continuous toolpaths, allowing complex geometries to be printed while maintaining their form. The methodology achieves three key objectives: (1) optimizing material distribution through force-informed design to create tension–compression combined structures, (2) enabling rapid fabrication of complex geometries that would be impractical with traditional construction methods that require formwork, and (3) developing a modular approach that overcomes build volume limitations through strategic segmentation and post-tensioning assembly techniques. This workflow is validated by designing, printing, and constructing a three-meter beam prototype, demonstrating how the comprehensive digital design approach results in a physical structure through additive manufacturing and construction.

1. Introduction

Additive manufacturing allows for geometric flexibility through controlled material placement [1–3]. Specifically, providing the opportunity for the fabrication of complex geometries digitally developed that were previously difficult to manufacture, while reducing the need for additional materials [4,5]. However, printable geometry is constrained by technical limitations, for example, maximum overhang angles and build volume restrictions [6–9]. This work presents an integrated computational design and advanced manufacturing workflow for constructing and assembling 3D concrete printed structures, designed and analyzed using a form-finding approach with embedding complex geometries.

1.1. 3D concrete printing structures

3D Concrete Printing (3DCP) enables the exploration of new possibilities for complex forms that would be impractical with traditional construction methods. It can be used for the fabrication of large-scale structures, enabling a wide variety of applications in architectural and civil engineering applications [10–14]. For example, 3DCP has been used to fabricate customized design of columns [15], unreinforced concrete bridges with optimized arch geometries [16], beams with toolpath-based designs [17], beam with supporting cables [18], and code-compliant pedestrian bridges [19], bridge design with topology optimization [20].

While the above 3DCP structures have explored various approaches to structural requirements and scale, existing methodologies face fundamental limitations. Compression-only designs achieve structural efficiency but require extensive support systems, and geometric constraints to maintain pure compression paths. Post-tensioned and reinforced approaches address tensile forces but typically rely on solid cross-sections or conventional reinforcement strategies, limiting material optimization potential and increasing fabrication complexity.

This research presents an approach generates structurally optimized forms, explicitly incorporating tensile forces as design parameters

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rather than constraints to be avoided or passively resisted. Second, the implementation of periodic anticlastic geometry for materialization creates a dual system of solid volumes and voids, achieving material reduction compared to solid cross-sections while maintaining structural capacity through optimized surface curvatures. This differs fundamentally from topology optimization approaches that typically result in irregular geometries challenging for layer-based fabrication. Third, the research introduces a segmentation and assembly strategy that extends 3DCP structures beyond printing bed limitations while maintaining structural continuity. The method ensures geometric compatibility through computationally designed interfaces where adjacent segments share identical layers of geometry. Application of post-tensioning serves dual purposes: providing tensile capacity for spanning and creating compressive preload at interfaces for force transfer, thereby unifying discrete printed elements without mechanical joints.

1.2. Implementation of anticlastic surfaces

Anticlastic surfaces exhibit negative Gaussian curvature, where principal curvatures at any point have opposite signs, one curves upward while the perpendicular curves downward, forming a saddle-point geometry. This geometric property distinguishes them from surfaces such as spheres or domes, where both curvatures bend in the same direction [21]. The resulting opposing curvatures create natural stiffening that resists deformation, enables efficient load distribution across the surface, and allows structures to achieve large spans with minimal material thickness [22]. The structural potential of anticlastic surfaces has been extensively explored in architectural practice, with pioneering work by Frei Otto demonstrating the application of minimal surface principles in large-scale projects, including the German Pavilion for Expo 1967 and the Munich Olympic Stadium [21,23]. Other notable architects and engineers have similarly leveraged these surfaces for lightweight structural applications, including Felix Candela's hyperbolic paraboloid shell structures [24] and Sergio Musmeci's innovative long span structures [25].

Within the broader category of anticlastic surfaces, minimal surfaces represent a special subset with distinct mathematical properties. At each point on a minimal surface, considering k_1 and k_2 as the principal curvatures, the mean curvature equals zero ($H = (k_1 + k_2)/2 = 0$) while the Gaussian curvature remains negative ($G = k_1 \cdot k_2 < 0$) [26]. This zero mean curvature condition indicates that the surface area is locally minimized for given boundary conditions.

Minimal surfaces naturally occur in physical phenomena such as soap films stretched across wire frames, where surface tension drives the formation toward minimum area configurations. This mathematical property translates to structural applications, as minimal surfaces inherently balance tensile forces uniformly across their extent. Among minimal surfaces, Triply Periodic Minimal Surfaces (TPMS) constitute a particularly class. These mathematically defined continuous surfaces extend periodically in three orthogonal directions and exhibit enhanced mechanical performance through their periodic nature [27–32]. This research explores periodic anticlastic geometry transformed from TPMS as a viable structural form at the construction scale, achieving a more uniform distribution of forces while maintaining structural stiffness. By mathematically transforming these surfaces through parametric equations, the workflow enables the printing of structures that would otherwise be impossible with traditional concrete printing approaches. This adaptation process ensures that the geometry maintains its structural form while conforming to the practical limitations of layer-by-layer extrusion in 3DCP.

1.3. Methods of structural form-finding

Structural form-finding is the set of methods that determine optimal structural geometries using equilibrium principles, ranging from historical physical techniques to contemporary computational approaches. The classical physical methods, implemented by architects such as Antoni Gaudíand Frei Otto, utilize hanging chain models and fabric membranes to determine compression-only forms [21,22]. These methods provide an intuitive understanding of force flow and generate naturally optimized geometries. However, they present limitations: physical models are difficult to scale, challenging to modify iteratively, and restricted to specific structural types, primarily compression-only systems. Additionally, translating physical models into digital construction documents requires extensive surveying and reconstruction.

Computational form-finding methods have emerged to address these limitations. The Force Density Method (FDM), developed by Schek [33], represents one of the earliest computational approaches, solving for nodal positions through force-to-length ratios. While FDM offers computational efficiency and ease of implementation, it provides limited intuitive control over resulting forms with complex loading conditions. Dynamic Relaxation methods [34] simulate physical behavior through iterative equilibrium calculations, offering flexibility in handling various structural types. However, these methods require careful parameter tuning and can be computationally intensive for complex geometries. Thrust Network Analysis (TNA) [35] extends 2D graphic statics to three-dimensional funicular structures, offering intuitive control over compression networks. TNA excels in designing masonry structures but remains limited to compression-only systems, and requires additional strategies for incorporating tension elements. Particle-spring systems [36] provide interactive real-time form exploration but experience discretization errors. Separately, genetic algorithms [37] enable extensive design space exploration through evolutionary optimization, though convergence to optimal solutions requires many iterations and provides limited insight into the relationship between form and forces.

Polyhedral Graphic Statics (PGS) addresses several limitations of existing methods by extending graphic statics principles to three-dimensional polyhedral frameworks [38,39]. The method provides direct, intuitive control over force magnitudes through geometric manipulation of reciprocal diagrams, enabling designers to understand and modify force distributions precisely.

By modifying the force diagram to maintain uniform force magnitudes, the material can be strategically allocated, for example, placing concrete elements in compression zones and steel cables in tension regions to span structures. This force-informed approach also guides the placement of anticlastic geometries along force paths through identified structural cells. However, implementing such anticlastic geometry at the construction scale through 3DCP presents manufacturing challenges, particularly due to horizontal surfaces and unsupported overhangs that exceed typical printing constraints [27]. To address these limitations, this work develops a framework that employs mathematical transformations, optimization processes to minimize local overhang angles, and generate continuous toolpaths. Furthermore, to overcome build volume constraints, the structure is segmented into modules for fabrication and assembly, enabling the realization of larger-scale structures while maintaining geometric complexity.

1.4. Organization of paper

This paper is structured as follows: Section 2 presents the methodological framework, beginning with PGS-based structural form-finding and algebraic edge constraints, followed by geometric materialization through volumetric modeling and anticlastic surface integration, and concluding with fabrication optimization including overhang analysis and toolpath generation. Section 3 validates the workflow through a three-meter beam prototype, demonstrating the form-finding process, comparative TPMS analysis, geometric optimization algorithms, and toolpath development strategies. Section 4 details the physical implementation, documenting the 2K printing system, material specifications, fabrication parameters, and post-tensioned assembly methodology that transforms discrete printed segments into a unified structural system. Section 5 concludes with key contributions of the research,

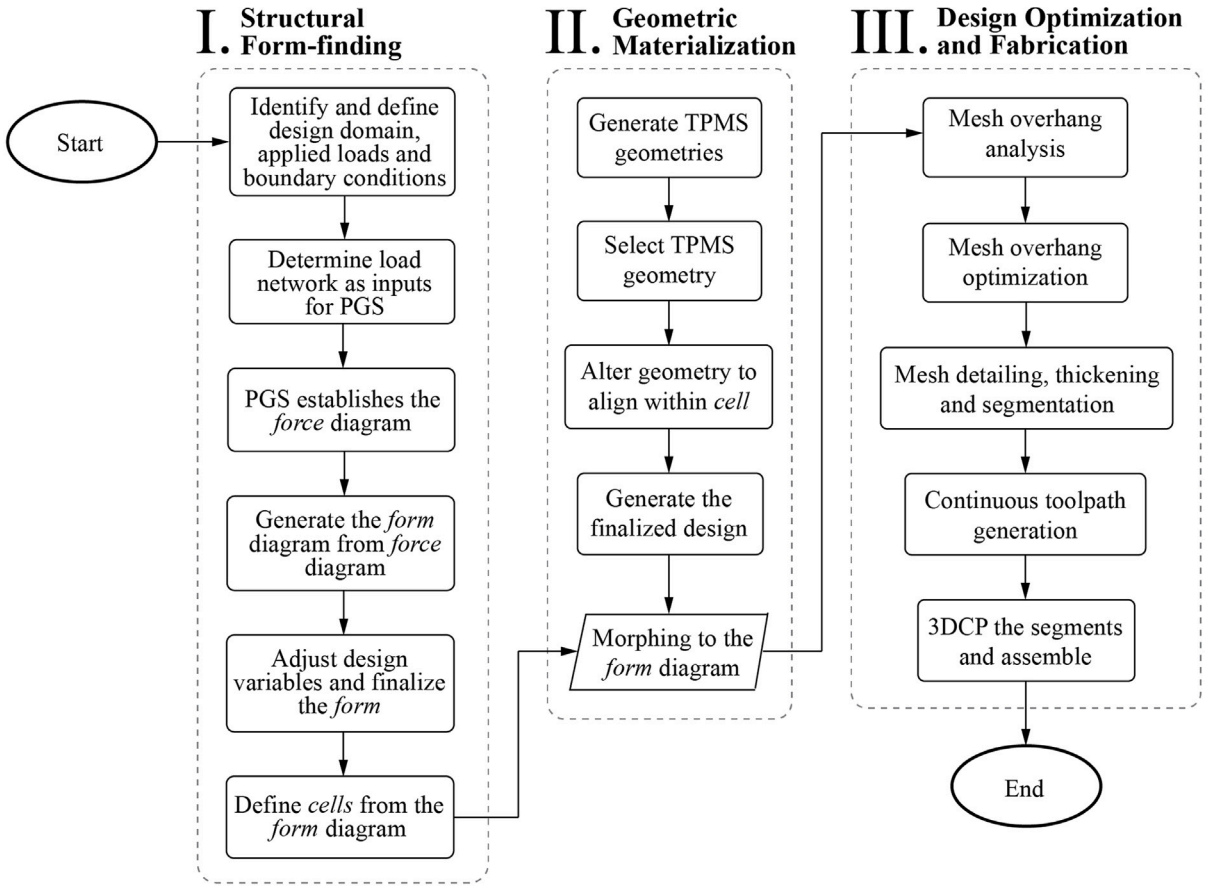


Fig. 1. Workflow consists of a three-phase process: (I) Form-Finding, establishing reciprocal form-force relationships; (II) Geometric Materialization, embedding anticlastic geometry within the form; and (III) Optimization and Fabrication, iteratively refining geometry under printability constraints.

including the successful integration of periodic anticlastic geometry for material optimization, and the validated fabrication strategy for large-scale 3DCP, and identifies future directions for scaling this approach to larger architectural applications, alternative material systems, and enhanced post-tensioning optimization strategies.

2. Digital-to-physical workflow

The overall design-to-fabrication workflow is presented in Fig. 1. The development is divided into three major phases: structural form-finding, geometric materialization, and design optimization and fabrication. The process begins with PGS-based form-finding, which includes defining boundary conditions, establishing force and form diagrams, and extracting cellular units. After a form is established, material is placed considering the embedment of the anticlastic surface by identifying applicable geometry, where design variables are identified and iteratively refined before morphing to the form diagram. The final phase accounts for fabrication constraints through mesh overhang analysis and optimization, resulting in modular structure with detailed meshes and toolpaths, ready to be printed and assembled. Each phase is discussed in the following sections.

2.1. Form-finding through graphic statics

The structural form-finding using Graphic Statics begins with defining boundary conditions, including applied loading and support conditions, as well as their respective location. Followed by the load network and mapping the primary force paths through the structure. The network identifies the point-loads of force application, visualizes the magnitude and direction of forces, and establishes the basic topology of the design. To illustrate the method in practice, consider a simplified example (Fig. 2) that contains components in systems with tensile and compressive forces. The initial form diagram in Fig. 2(a) shows a triangulated structure with one applied load and two reaction forces at supports. Each structural element in the form diagram has a corresponding representation in the force diagram. The force diagram Fig. 2(b) is constructed by arranging force vectors head-to-tail, forming a closed polygon that ensures force equilibrium. The magnitude and direction of forces are represented by the length and orientation of edges in the force diagram, where blue indicates compression and red indicates tension members. The reciprocal relationship between form (Γ) and force ($\Gamma^{\dagger}$) diagrams is established through a one-to-one correspondence of their geometric elements: vertices ($v = 4$) in Γ correspond to faces ($f^{\dagger} = 4$) in $\Gamma^{\dagger}$, edges ($e = 8$) in Γ map to edges ($e^{\dagger} = 8$) in $\Gamma^{\dagger}$, and faces ($f = 5$) in Γ transform to vertices ($v^{\dagger} = 5$) in $\Gamma^{\dagger}$. This duality enables direct manipulation of force distributions while maintaining structural equilibrium, as any modification in one diagram results in corresponding adjustments in the other.

The dual relationship between form and force can be further demonstrated through the application of Minkowski sum operations [40], which provide a mathematical framework for establishing the connection. In the context of graphic statics, the Minkowski sum represents the set of all possible vectors obtained by adding elements from each set, effectively mapping the transfer of forces between the two diagrams. As illustrated in Fig. 2(c), this transformation process from form to force enables verification of structural equilibrium by ensuring that the vectorial sum of forces at each joint equals zero. The progression of the transformation is demonstrated through varying values of parameter ψ, showing the adaptive nature of the form-force relationship and its impact on the different structural configurations.

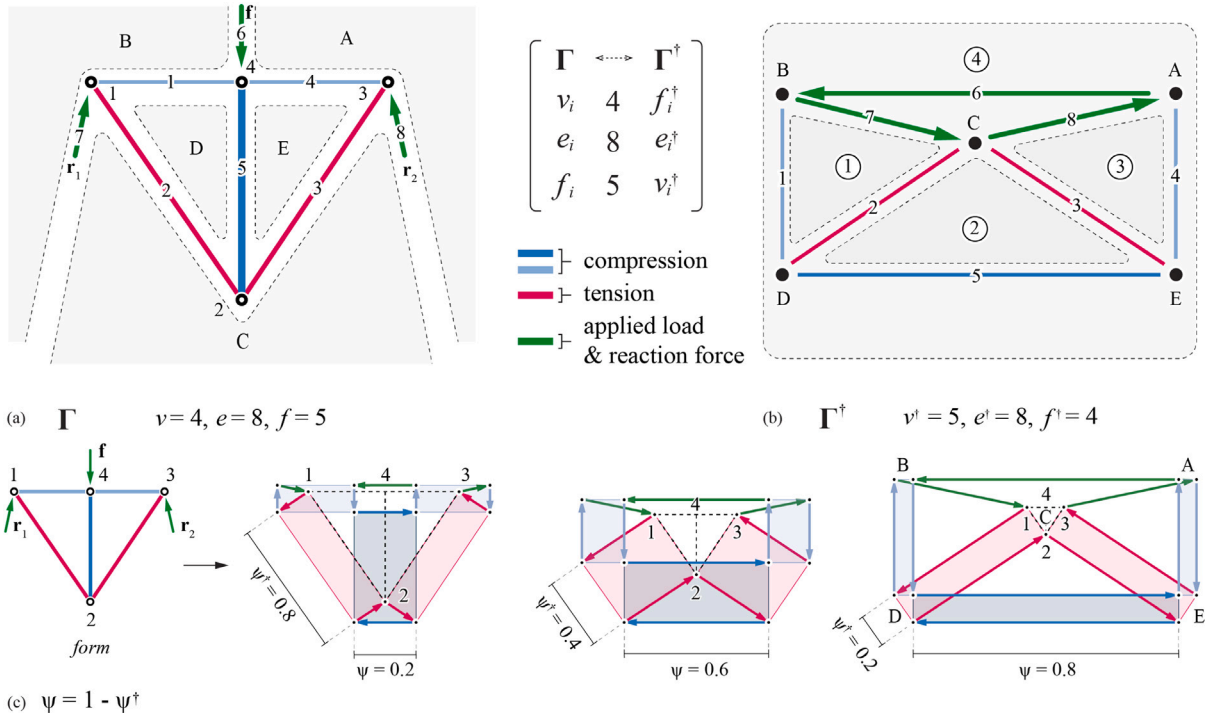


Fig. 2. (a) Initial form diagram showing vertices-v, edges-e, and faces-f, where $v = 4$, $e = 8$, and $f = 5$; (b) corresponding force diagram, where $f^{\dagger} = 4$, $e^{\dagger} = 8$, and $v^{\dagger} = 5$; (c) Minkowski sum transformation sequence illustrating the continuous mapping between form and force diagrams through parameter ψ (ranging from 0 to 1).

When applying the method in three dimensions, PGS introduces an additional geometric element *cell* in both form and force diagram while preserving the fundamental principles of reciprocity. The *cell* in the form diagram is extracted in Phase II and used to embed geometries within its bounds.

2.2. Geometric materialization

Recent structures designed by PGS have explored various material applications [41–43]. While previous approaches focused on translating form diagram edges(e) into materialized solid members or faces(f) into planar sheets, this workflow utilizes the form diagram cells(c) as spatial hulls, providing a force informed boundary for incorporating anticlastic geometries, enabling the integration of structural form-finding through PGS with lightweight surface systems. Schwarz, Schoen, and others [44] described these minimal surfaces through patch-wise construction and analytical methods, focusing on their mathematical properties as minimal surfaces with zero mean curvature at every point. Approximating TPMS geometry trigonometric functions allows for practical control over porosity and volume through parametric equations, providing a computational method to generate structures that mimic minimal surfaces and can be combined with other true SDFs. These periodic minimal surfaces have been applied beyond their mathematical origins to find applications across multiple scientific and engineering disciplines, with their geometric principles and structural properties serving a variety of purposes in mathematics, physics, materials science, engineering, and architecture [45–48].

The generation of periodic minimal surface geometries presents challenges in traditional Computer-Aided Design (CAD) systems. While conventional boundary representation (BRep) methods [49] define geometry through vertices, edges, and surfaces, they prove inadequate for handling the complex mathematical surfaces inherent in lattice geometries. Function representation (FRep) [50] offers a more suitable approach by defining geometry through the direct implementation of mathematical functions rather than discrete geometric elements. Volumetric modeling (VM) [51] extends this capability by representing objects as functions to define space as a scalar field rather than surface meshes, providing precise control over complex geometric formations. Implementation of SDF with VM and FRep [52,53] enables practical realization of these computational approaches. SDFs define geometry as a mathematical function that directly embodies distance to a surface throughout space. The continuous scalar field exists independently of any specific points where it might be evaluated. The relationship is expressed mathematically as $v = f(x, y, z)$, where v represents the distance from point (x, y, z) to the object defined by f. The object's surface is defined where $v = 0$, with positive values indicating external points and negative values denoting internal points. Such mathematical framework facilitates the generation of various TPMS configurations, although TPMS functions are actually signed scalar fields rather than SDFs—they identify surfaces at zero values but do not preserve the distance property, meaning their values do not represent actual distances from the surface. Different TPMS geometries offer distinct geometric characteristics suited for various applications (Fig. 3). The Primitive surface (Schwarz-P) features simple cubic symmetry, serving as a fundamental template for understanding more complex TPMS formations. Like the Primitive, the Gyroid surface, discovered by Schoen [54] also exhibits continuous curvature, but is further distinguished by its complete absence of straight lines. The Gyroid's unique morphology and balanced channel networks make it suitable for fluid dynamics applications such as heat exchangers and flow systems [55–57]. The Diamond surface (Schwarz-D) characterized by its diamond cubic symmetry and network of embedded straight lines (Fig. 4), making it particularly advantageous for PGS-based structural applications. Additional variants such as Neovius, F-Koch, and Lidinoid surfaces demonstrate the extensive range of TPMS configurations, each offering unique spatial arrangements and varying degrees of topological complexity that can be tailored to various requirements.

2.3. Design optimization and fabrication

The third phase of the workflow addresses fabrication constraints and optimization requirements to ensure the successful materialization

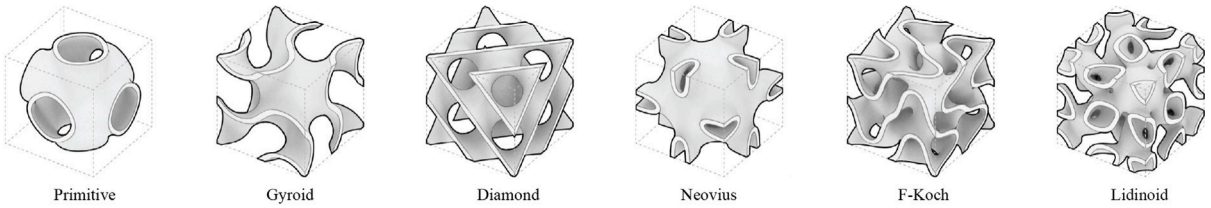


Fig. 3. Fundamental types of TPMS. From left to right: Schwarz-P (primitive) surface with simple cubic symmetry, Gyroid with continuous curvature, Diamond (Schwarz-D), Neovius, F-Koch and Lidinoid demonstrating complex periodic spatial configurations. Each surface maintains zero mean curvature.

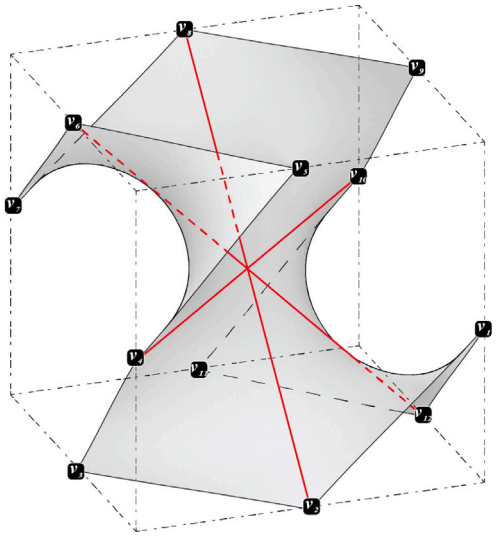


Fig. 4. The network of straight lines (red) contained in the Schwarz-D unit cell.

of the computationally generated designs. Following the geometric materialization and morphing of anticlastic structures to the form diagram, the workflow progresses through several fabrication-oriented steps. As illustrated in Fig. 1, the phase begins with a mesh overhang analysis to evaluate printability constraints related to self-supporting angles necessary in additive manufacturing processes. The identified overhangs inform the subsequent mesh optimization stage, where geometry is adjusted to minimize or eliminate problematic areas while preserving the structural form established in the previous phases.

After optimization, the mesh undergoes detailing, thickening, and segmentation processes to prepare it for manufacturing. The thickening transforms the mathematical surface into a printable shell with appropriate material thickness, while segmentation divides the overall structure into manageable modules suited to the printing volume constraints of available fabrication equipment. Continuous toolpath generation follows, converting the geometric representation into machine instructions that define the precise movement patterns for material deposition during printing. The workflow culminates with the 3DCP of individual segments and their subsequent assembly into the final structure.

This systematic approach to fabrication preparation ensures that the complexities of anticlastic geometries embedded within the PGS-informed cells can be physically realized. The specific implementation and validation of this phase is detailed through a three-meter beam prototype presented in the following sections.

3. Workflow validation and application

To demonstrate the design-to-fabrication workflow, a three-meter beam was designed, fabricated, and assembled. Fig. 5 illustrates the comprehensive implementation process through the same three phases in the previous section. The first phase employs PGS-based form-finding to establish the optimal spatial force distribution within the structure. In the second phase, diamond-based TPMS units are generated and optimized from systematic geometric transformations to achieve the designated printable anticlastic form while maintaining structural form. The final phase addresses fabrication requirements through multiple steps: detailed overhang analysis to ensure printability, incorporating cable integration and reinforcement placement, and development of an optimized continuous toolpath. The structure is ultimately resolved into discrete, printable segments, each carefully optimized for 3DCP constraints. This prototype demonstrates the successful translation of complex computational design principles into physically realizable components, validating the proposed workflow's effectiveness from conceptual design through final assembly.

3.1. PGS in structural form-finding

The principles of graphic statics can be extended from two to three dimensions to achieve spatial force distribution systems. Fig. 6(a) illustrates this transformation process, where a 2D force line-based diagram is extruded toward a point to generate a 3D force polyhedron. This dimensional expansion fundamentally changes how forces are represented, transitioning from edge-based representations in 2D to face-based representations in 3D. The force polyhedron, defined by parameters of width, length, and height, creates intersecting cells that represent tensile and compressive forces within the diagram. These cells are differentiated by their mechanical behavior: compression cells are visualized in blue, while tension cells are shown in red. The corresponding form diagram, shown in Fig. 6(b), manifests the spatial reciprocal relationship between force and form. The structural topology is represented through a series of nodes and members, with external loads network (f_i) and support reactions (r_i) indicated by vertical arrows.

Additionally, the Minkowski sum operation extends naturally to three-dimensional configurations. Fig. 7 illustrates this spatial transformation through the parameter ψ, where the relationship $\psi = 1 - \psi'$ governs the reciprocal mapping between form and force diagrams. The transformation sequence demonstrates how force patterns evolve from distributed states ($\psi = 0.8$) to concentrated configurations ($\psi = 0.2$), while continuously maintaining structural equilibrium between compression and tension elements. This parameterized geometric transformation enables dynamic transformation between form and force diagrams while providing mathematical validation of reciprocal relationships in three-dimensional space.

3.1.1. Algebraic form-finding in PGS

The previous operation in PGS relies purely on geometric constructions, but an algebraic approach offers enhanced computational capabilities for form-finding. This algebraic PGS method addresses the computational limitations inherent in previous procedural modeling methods, particularly for rapidly generating complex geometries [58–60]. The core principle is mathematically expressed through the perpendicularity condition between force diagram faces and their corresponding form diagram edges. While this algebraic formulation

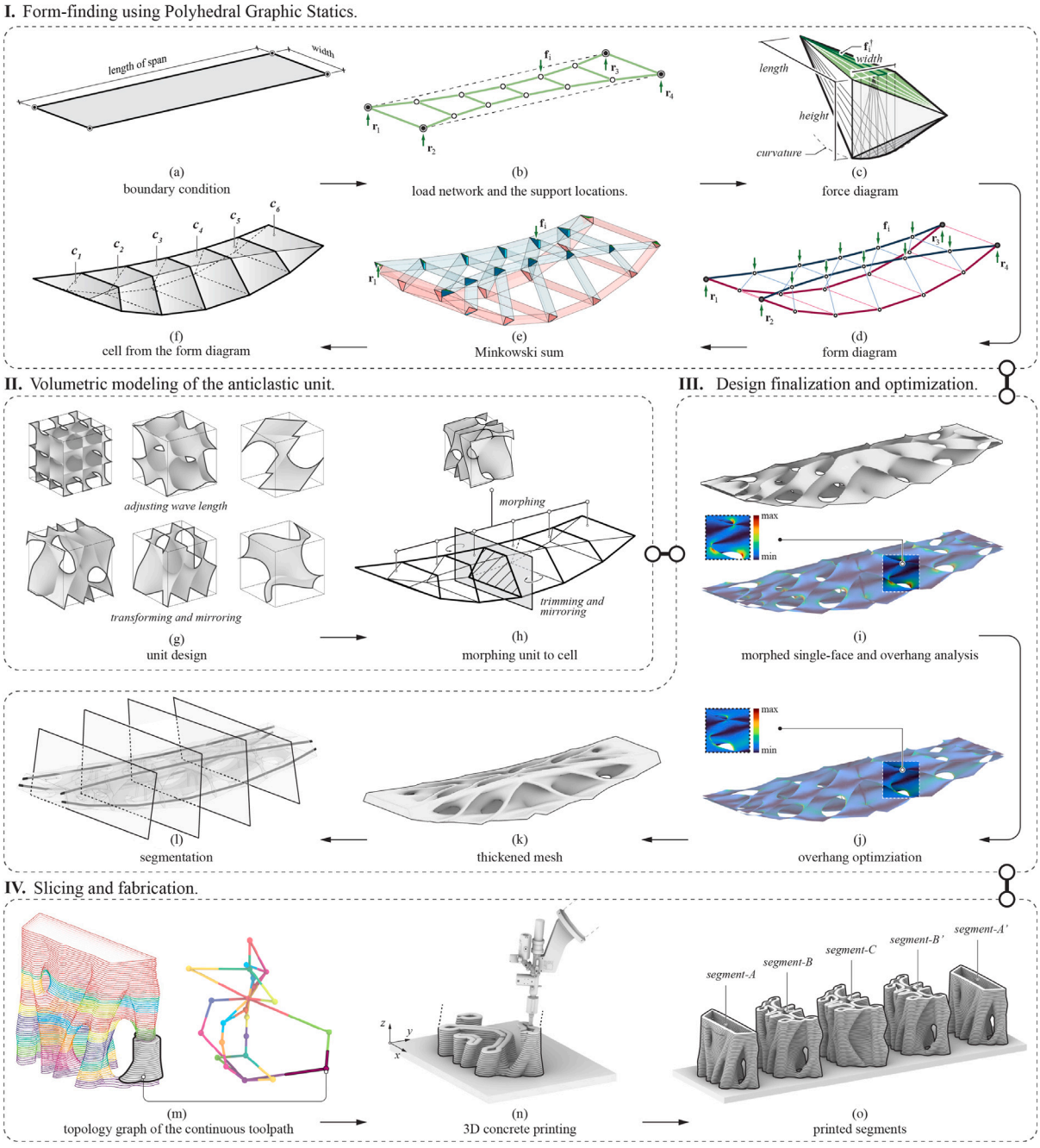


Fig. 5. (a) Setup the boundary condition; (b) define the load network and the support locations; (c) establish the force diagram with geometric parameters; (d) form diagram is generated based on the force diagram; (e) Minkowski sum describes the force distributions; (f) cells from form diagram; (g) diamond is selected to have multiple geometric operations to generate the final unit geometry; (h) the final determined unit geometry will be morphed to cells and will be trimmed and mirrored; (i) the overhang analysis is implemented to identify the problematic areas; (j) the geometry optimization algorithm is applied to minimize the overhang angles for additive manufacturing; (k) thickened mesh is generated with embedding conduit for cables and end volume for reinforcements; (l) generating the cutting planes to adjust the angle to avoid overhangs; (m) topology graph of the continuous toolpath shows the printing sequence and connections of each part; (n) extruding the geometry based on the sliced toolpath; (o) prototype is printed with five pieces.

enables greater flexibility in exploring design solutions, its current implementation is restricted to convex geometries as input. To overcome this limitation and achieve combined compression–tension structures, algebraic edge constraints are introduced to handle intersected faces effectively.

The mathematical framework begins with the definition of spatial coordinates through diagonal matrices $\mathbf{N}x$, $\mathbf{N}y$, and $\mathbf{N}z$ $[f_{intl} \times f_{intl}]$, which represent the directional components of internal face normal vectors. The vector $\mathbf{q}$ corresponds to the *force densities*, which serve as scaling factors governing the edge lengths in the dual. These matrices,

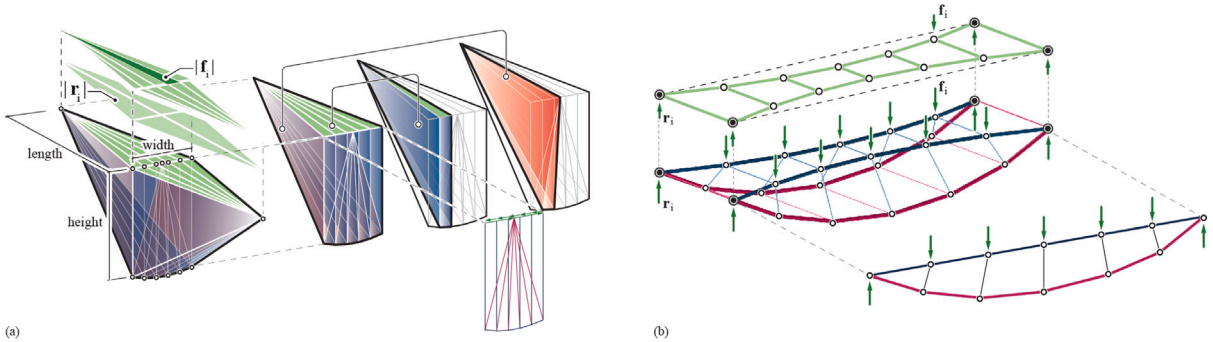


Fig. 6. Geometric representation of 3D graphic statics. (a) Force polyhedron generated through point extrusion, showing the transformation of planar forces into spatial cells. The polyhedron's geometry is controlled by fundamental parameters: width, length, and height; (b) reciprocal form diagram illustrating the spatial structural configuration with compression and tension members.

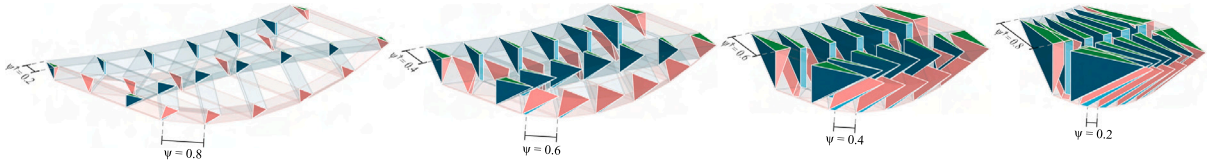


Fig. 7. Minkowski sum transformation in 3D is controlled by parameter ψ, where $\psi = 1 - \psi'$. The sequence shows progressive consolidation of force patterns from (a) $\psi = 0.8$ exhibiting distributed configuration, through (b) $\psi = 0.6$ and (c) $\psi = 0.4$, to (d) $\psi = 0.2$ showing concentrated force distribution.

combined with the connectivity matrix $\mathbf{C}_{e_{intl} \times f_{intl}}$ that establishes relationships between faces and edges, form the basis of the equilibrium equations:

$$\begin{aligned} \mathbf{C}_{e_{intl} \times f_{intl}} \mathbf{N}_x \mathbf{q} &= \mathbf{0} \\ \mathbf{C}_{e_{intl} \times f_{intl}} \mathbf{N}_y \mathbf{q} &= \mathbf{0} \\ \mathbf{C}_{e_{intl} \times f_{intl}} \mathbf{N}_z \mathbf{q} &= \mathbf{0} \end{aligned} \tag{1}$$

These equations are consolidated into a unified equilibrium matrix $\mathbf{A}$ $[3e_{intl} \times f_{intl}]$:

$$\mathbf{A} = \begin{pmatrix} \mathbf{C}_{e_{intl} \times f_{intl}} \mathbf{N}_x \\ \hline \mathbf{C}_{e_{intl} \times f_{intl}} \mathbf{N}_y \\ \hline \mathbf{C}_{e_{intl} \times f_{intl}} \mathbf{N}_z \end{pmatrix} \tag{2}$$

To achieve the intersected face condition through geometric controlling, edge constraints are introduced through additional equations. For each constrained edge $e_i^{\dagger}$, a target length l is specified through the constraint equation:

$$\mathbf{b}_i^{\mathrm{T}} \mathbf{q} = l \tag{3}$$

where $\mathbf{b}_i$ represents a $[f_{int} \times 1]$ column vector with unit value at the constrained edge index. Multiple edge constraints are assembled into a matrix form:

$$\mathbf{B}\mathbf{q} = \mathbf{l} \tag{4}$$

The edge constraint matrix $\mathbf{B}$ has dimensions $[b \times f_{int}]$, where each row is denoted by the row vector $\mathbf{b}_i^{\mathrm{T}}$. The target lengths are contained in the column vector $\mathbf{l}$ of dimensions $[b \times 1]$.

The complete system integrates both equilibrium requirements and geometric constraints into a non-homogeneous constrained linear equation:

$$\mathbf{M}\mathbf{q} = \mathbf{t} \tag{5}$$

where $\mathbf{M}$ represents the constrained equilibrium matrix, column vector $\mathbf{t}$ is obtained by vertically stacking the $[2e_{intl} \times 1]$ zero vector and the vector $\mathbf{l}$. Each solution vector $\mathbf{q}$ contains a set of edge lengths that, when applied to the form diagram, simultaneously satisfies both the closing conditions and all specified constraints. This formulation enables precise control over geometry while maintaining equilibrium conditions.

Fig. 8 demonstrates the systematic transformation of force and form diagrams through algebraic edge constraints. Throughout the sequence, face colors consistently represent force directions: blue faces indicate negative normal vectors (compression forces), while red faces denote positive normal vectors (tension forces). The transformation begins with basic force polyhedrons (Fig. 8a), where faces f_1, f_2, and f_3 establish the initial geometric configuration. The normal vectors n_{f_i} of these faces determine force directions, while force algebraic edge constraints (Fig. 8b) are introduced through the adjustment of lengths l_1 and l_2 ($l_1, l_2 > 0$), controlling the relationships between compression and tension zones. As the transformation progresses (Fig. 8c–d), face f_1 maintains its tension status with the same normal direction, while face f_2 transitions from tension to compression as its normal vector flips to the negative direction, resulting in modified constraint conditions ($l_1 < 0, l_2 > 0$). This manipulation drives the evolution of the form diagram (Fig. 8e), which progresses through multiple phases. Starting from an all-tension configuration in Phase 1, the structure evolves through stages 1a-1d, demonstrating the transition from tension-dominant to mixed compression–tension states. Members e_1 and e_2 in the form diagram correspond directly to faces f_1 and f_2 in the force diagram. The transformation concludes with refined force diagrams (Fig. 8f–g) under adjusted constraints ($l_1 < 0, l_2 = 0$), culminating in the final structural configuration of Phase 2 (Fig. 8h).

3.1.2. Geometric optimization for tensile forces

Building upon the algebraic form-finding process, further geometric optimization is implemented to achieve uniform tensile forces throughout the structure. This optimization is accomplished through strategic manipulation of the force diagram's bottom curvature. Fig. 9 illustrates this transformation sequence, where each applied load face or edge is normalized to 1 kN to enable proportional force calculation across the system. The optimization process demonstrates the direct relationship between the force diagram's geometry and the resulting force distribution in structural members. In the initial configuration, the tension members (shown in red) exhibit non-uniform force magnitudes

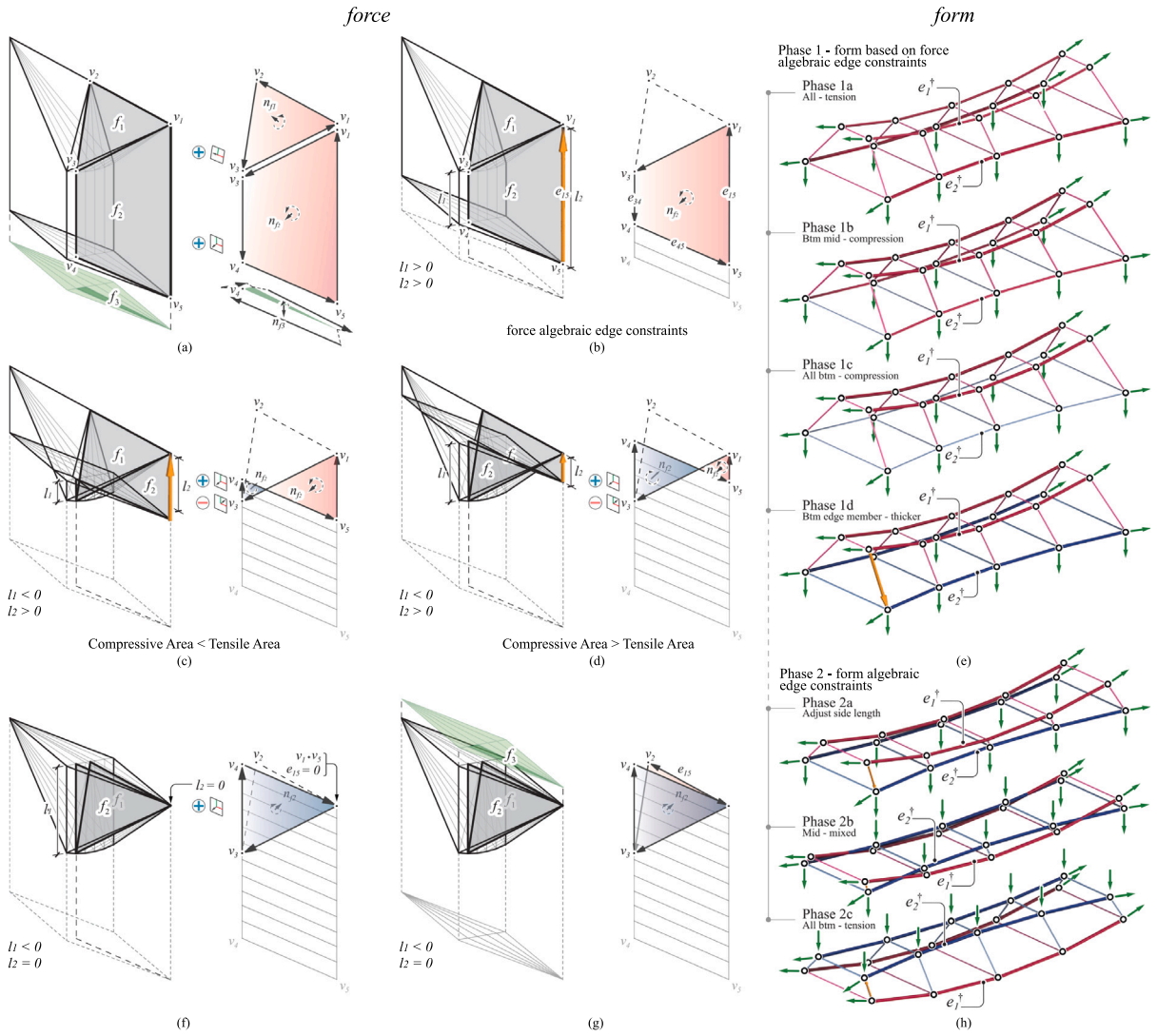


Fig. 8. (a) Initial force polyhedron with faces f_1, f_2, f_3 and its dual representation with normal vectors n_{f_i}; (b) identify algebraic edge constraints l_1, l_2 in the force diagram; (c–d) progressive transformation showing the relationship between compressive and tensile areas with varying edge length constraints ($l_1 < 0$, $l_2 > 0$); (e) sequential development of form diagrams through Phases $1_a - 1_d$, demonstrating the evolution from all-tension to mixed compression configurations with varying member thicknesses; (f–g) force diagram with finalized edge constraints ($l_1 < 0$, $l_2 = 0$); (h) Phase 2 form diagrams showing the progression from adjusted side lengths to compression at top and tension at bottom configuration.

ranging from 3.61 kN to 5.07 kN. To address this variation, a circular arc is introduced at the bottom of the force diagram. This geometric modification creates uniform face areas in 3D (or edge lengths in 2D), resulting in balanced force magnitudes of 3.39 kN across all tension members. This force uniformity is particularly crucial for 3D printed structures where consistent post-tensioning is essential. By ensuring uniform tension forces throughout the system, the design better accommodates directional variations in material properties inherent to the layered manufacturing process. Furthermore, this balanced force distribution helps prevent delamination between printed layers and minimizes stress concentrations at layer interfaces.

3.2. Materialization with periodic anticlastic geometry

Once the initial structure is designed through PGS, the next phase focuses on materializing the tension and compression load paths through periodic anticlastic geometry, rather than directly translating the edges of the form diagram as in traditional approaches. While TPMS offers significant geometric flexibility for creating anticlastic surfaces, its implementation must carefully consider fabrication constraints inherent to additive manufacturing processes, particularly in 3DCP applications. The critical challenge in translating these mathematical surfaces into printable structures lies in controlling overhangs, as unsupported material beyond certain angles can lead to print failure in 3DCP.

3.2.1. Comparative analysis of TPMS configurations for 3DCP

To ensure fabrication feasibility, a systematic analysis of potential overhang displacement is conducted. The structural modules are contained within 1 : 3.3 : 1.9 boxes, representing the same proportion of the beam segments that will be printed (Type A as in Fig. 16), where a recommended maximum displacement ratio D of 0.5 is established to maintain printability.

A comparative analysis of four TPMS configurations demonstrates varying degrees of printability based on overhang displacement D. (The overhang displacement D is defined as the projected offset distance divided by the layer height [61,62].) Fig. 10(a) presents the geometric characteristics of Diamond-a, Gyroid, Diamond-b, and Fischer–Koch modules, with their corresponding overhang analysis visualized through color mapping [62]. The analysis reveals variations in overhang distribution, where red regions indicate areas with surface normals exceeding critical angles from the vertical axis (typically beyond

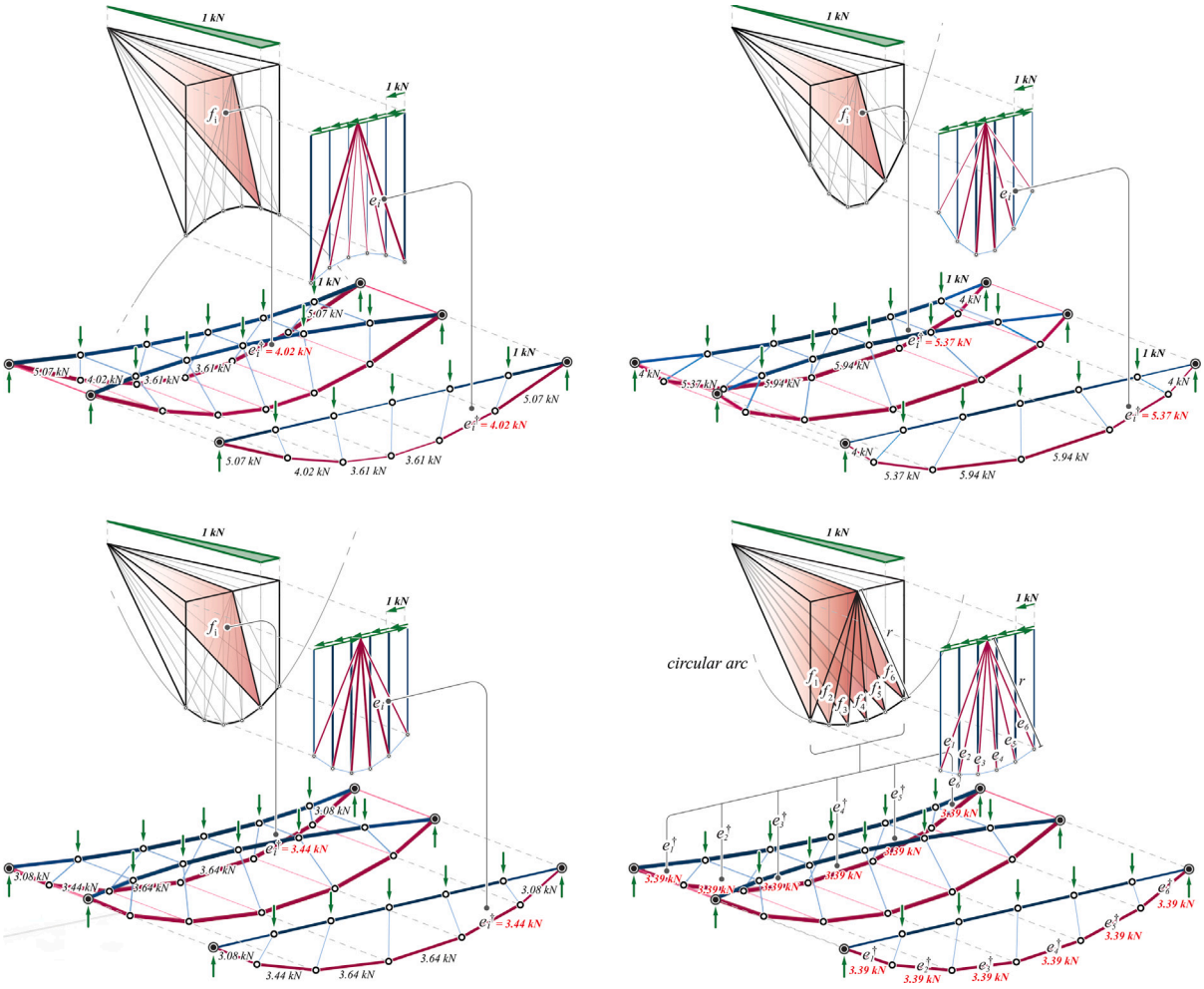


Fig. 9. Optimization of tension forces through bottom curvature modification in force diagram. Starting from varying forces (3.61 – 5.07 kN), the implementation of a circular arc achieves uniform tension (3.39 kN) under 1 kN applied loads. Force magnitudes in the form diagram correspond to edge lengths in 2D and the area of faces in 3D in the force diagram.

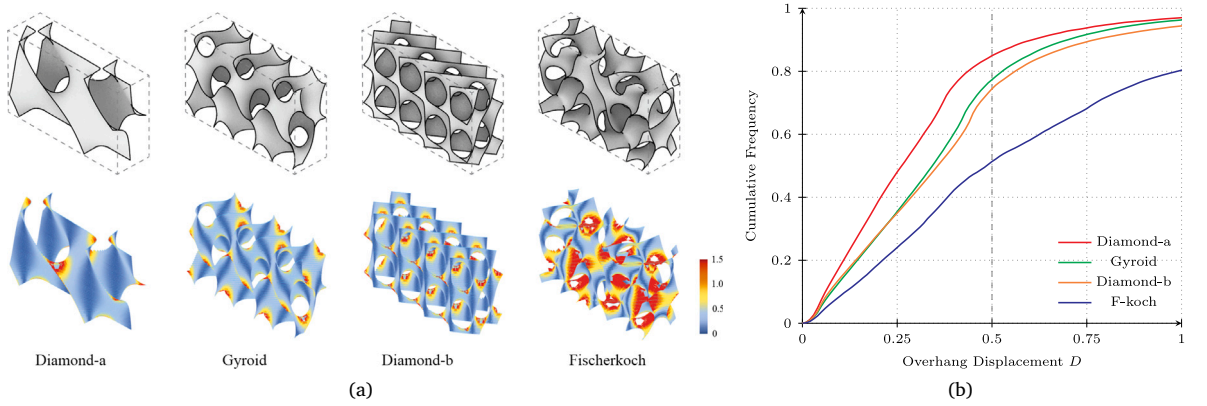


Fig. 10. (a) Overhang displacement D of the four adjusted TPMS modules; and (b) their cumulative frequency curve. 85% of the Diamond-a geometry has a D below 0.5.

45 degrees or an overhang displacement of 1), requiring printing support or geometric modification. The cumulative frequency distribution of overhang displacement, shown in Fig. 10(b), provides quantitative insights into the printability of each configuration. The Diamond-a geometry exhibits superior printing characteristics, with 85% of its surface area maintaining D values below 0.5, indicating favorable self-supporting properties. The Gyroid and Diamond-b configurations display similar distribution patterns, though with slightly higher overall overhang values. In contrast, the Fischer–Koch geometry shows significantly higher overhang displacements across its surface, as evidenced by its lower cumulative frequency curve, suggesting potential challenges in direct additive manufacturing.

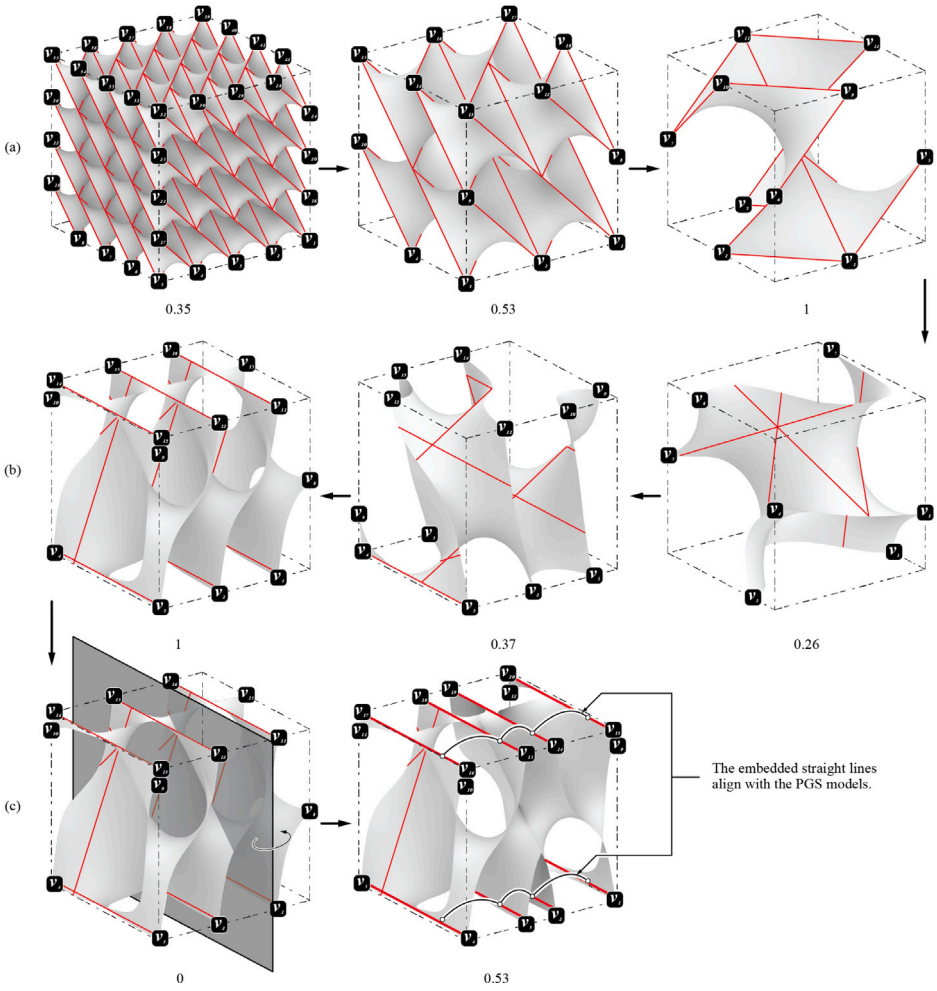


Fig. 11. Geometric transformation sequence of Diamond-A TPMS unit cell. (a) wavelength variations show pattern density evolution; (b) unit cell transformation for the alignment with the PGS model; (c) cutting and Mirroring determine the final unit shape.

This analysis confirms the Diamond-a configuration's feasibility for 3DCP, distinguished by its embedded straight lines that effectively contain compression and tension forces within the surface. Fig. 11 shows the geometric manipulation of the unit design for the further development. The initial transformation begins with wavelength adjustment Fig. 11(a), where the periodicity of the diamond unit is modified through values ranging from 0.2 to 1.0. This operation alters the density and distribution of the surface pattern, directly influencing the overall density of the resulting geometry. The progression demonstrates how wavelength modification affects the space of the minimal surface, transitioning from densely packed formations to more expansive surface configurations ideal for large-scale additive extrusion. The second phase implements matrix transformation operations Fig. 11(b), utilizing rotation matrix to orient the base geometry. This mathematical transformation enables precise alignment of the unit with the member forces direction in the form diagram, ensuring proper integration with the post-tensioning cables and feasibility of the printing. The cutting operation and mirroring Fig. 11(c) introduces strategic modifications to the base unit, creating more even loading conditions in the surface continuity. During unit cell arrangement, anticlastic geometry at interface locations is modified to maintain surface continuity and ensure smooth transitions across the entire structural system. The integer wavelength constraint ensures that the surface properties, including curvature and minimal surface characteristics, remain consistent at cell boundaries, preventing discontinuities that could compromise structural performance.

3.2.2. Embedding anticlastic geometry

After establishing the base unit geometry, the next phase involves its integration into the PGS-derived structural form. The process of applying this anticlastic geometry unit to the PGS model is summarized in Fig. 12. The form diagram is generated with six individual cells Fig. 12(b), labeled as c_i, which serve as the spatial hull for material distribution. These cells, varying in geometry to accommodate the structural curvature, establish the boundaries for integration. The transformation of anticlastic units into the defined cellular framework is achieved through a twisted box morphing Fig. 12(c). Each vertex of the unit is mapped to corresponding cell vertices, with a labeled vertex indicating the mapping sequence. This morphing process preserves the mathematical properties of the anticlastic surface while adapting to the prescribed structural geometry. Additionally, the repetitive cellular pattern maintains continuity across the span.

Initially, the single-surface anticlastic unit is morphed to align with the hexahedral cells of the form diagram, maintaining topological relationships as the section A-A' in Fig. 12(d) showed. This deformed geometry is then converted into a distance field, enabling precise control of surface offsets for 3DCP. VM's computational approach ensures consistent material deposition spacing, which is critical for structural bonding by allowing exact distance calculations throughout the volume. The geometric implementation strategically aligns the embedded straight lines of the anticlastic surface with the designed structure's primary force paths. Section B-B' in Fig. 12(e) illustrates how this alignment facilitates force transfer while accommodating post-tensioning cables (showing in black dots) through controlled dilation

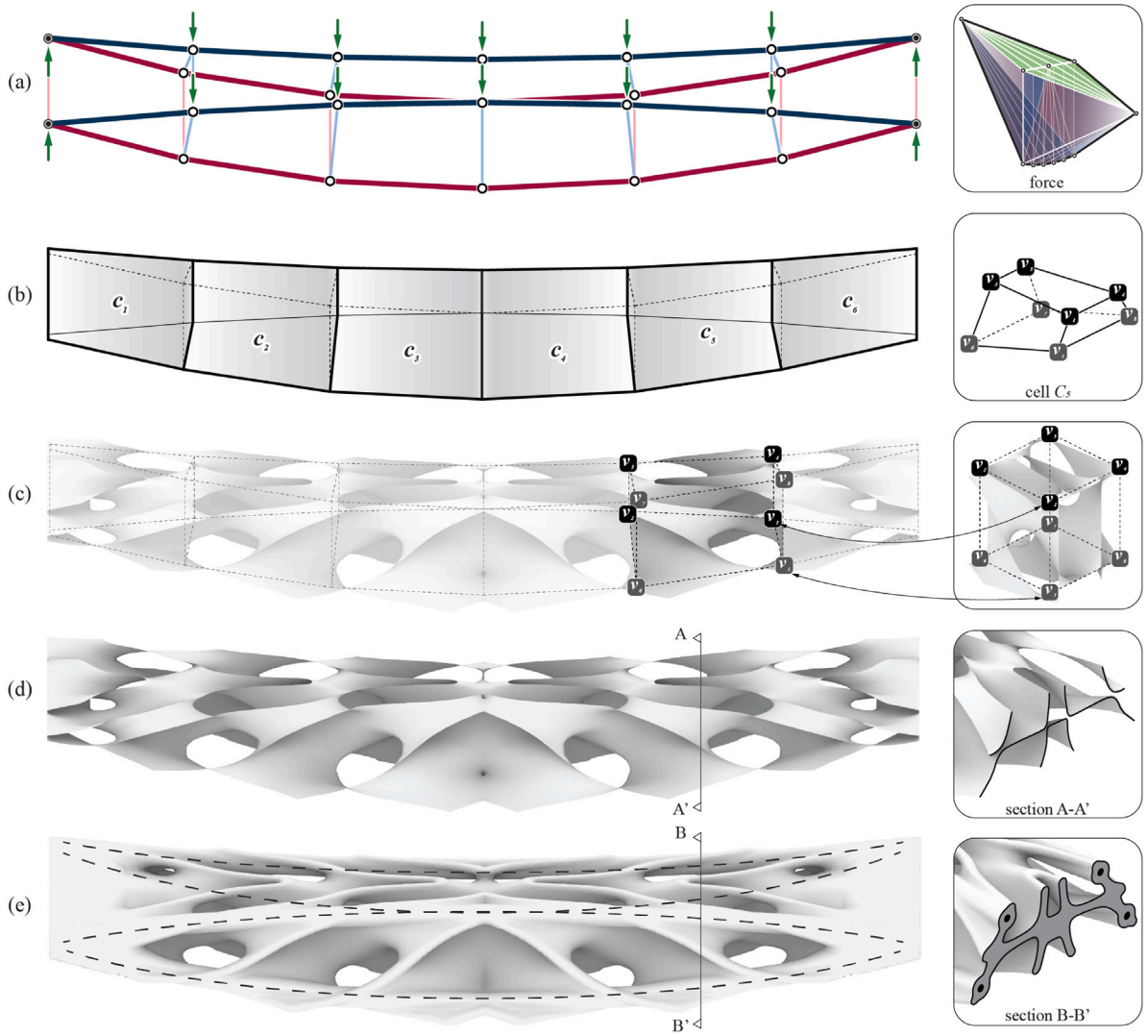


Fig. 12. Integrating PGS with the periodic anticlastic geometry. (a) Form diagram with compression (blue) and tension (red) members under vertical loads and corresponding force diagram; (b) cell segmentation of form diagram establishing volumetric boundaries; (c) implementation through twisted box morphing with vertex mapping correspondences; (d) section A-A' showing interconnected anticlastic surfaces with varying curvatures; (e) section B-B' illustrating thickening process through SDF operations and smooth transitions for integration of post-tensioning cables (dashed lines).

of print paths, indicated by the dashed lines. The VM approach excels in these modifications, enabling Boolean operations and smooth transitions between geometric features without the computational challenges typically encountered in mesh-based modeling. The section reveals how complex operations such as unions between different geometric elements are computed rather than constructed, allowing seamless integration of functional requirements while maintaining structural continuity and printability.

The feasibility of the developed steps was tested through small-scale prototyping using Polylactic Acid (PLA) material before applying it to large-scale 3DCP. Fig. 13 demonstrates the versatility of the implementation through four distinct configurations that were analyzed in the previous overhang chart, each manufactured using a desktop 3D printer. The reduced scale and PLA material properties enabled direct printing of complex geometries without additional support structures, as the material's rapid cooling characteristics better accommodate overhanging features compared to concrete printing. The prototypes, with a length of 30 cm each, showcase various configurations: Fig. 13(a) illustrates the Diamond-a geometry with its characteristic perpendicular intersections, which is selected for full-scale 3DCP. Fig. 13(b) demonstrates the Gyroid pattern with its continuous curved surfaces, Fig. 13(c) presents the Diamond-b variation featuring more complex spatial arrangements, and Fig. 13(d) shows the Fischer–Koch configuration with its unique periodic patterns. Each model consists of five segments that align with the segmentation strategy developed for the concrete prototype.

3.2.3. *Overhang optimization for 3DCP*

To further ensure printability, the optimization of the morphed structure focuses on controlling overhang angles is implemented. Based on the normal-driven shape analogy method developed by Liu and Jacobson [63], the optimization process constrains surface normals relative to the printing direction while preserving essential geometric features. Fig. 14 demonstrates this process, where Fig. 14(a) shows the initial geometry with problematic overhangs (in red) and Fig. 14(b) presents the optimized configuration with controlled normal angles.



Fig. 13. Small-scale PLA prototypes demonstrating different TPMS configurations. 1 cm bar shows the relevant scale. (a) Diamond-a has perpendicular surface intersections; (b) Gyroid features continuous curved surfaces; (c) Diamond-b shows complex spatial patterns; (d) Fischer–Koch configuration.

The optimization algorithm (Algorithm 1) implements printability constraints through iterative normal vector adjustment. The process begins with vertex normal computation and enforces a maximum overhang angle constraint relative to the printing plane. First, for each vertex, the algorithm computes the angle between its normal vector and a reference vector linked to the horizontal plane. When this angle exceeds the threshold, the normal is adjusted to maintain printability while minimizing distortion. Second, the mesh vertices are updated through a quadratic optimization that preserves the topology while conforming to the modified normal directions. Boundary vertices remain fixed to maintain the overall structure. The optimization continues until convergence or the maximum iteration count is reached. The implementation uses cotangent weights for mesh Laplacian computation and includes regularization to balance normal alignment with mesh smoothness. Surface normal distribution is visualized through color mapping, where lighter regions indicate steeper angles requiring optimization, and darker regions represent printer-compatible orientations.

4. Digital fabrication and construction

The fabrication and construction methodology follows a systematic process consisting of four interconnected phases to address the inherent

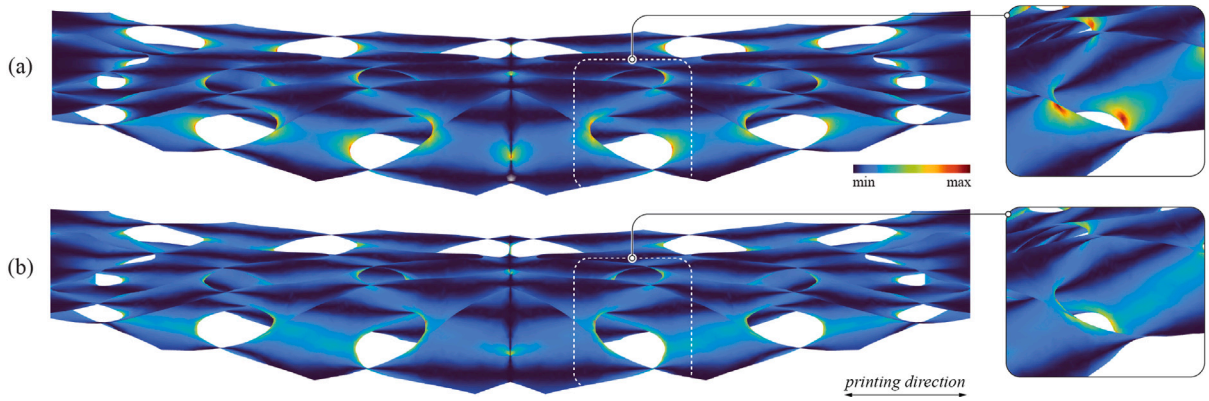


Fig. 14. Overhang angle optimization based on normal-driven shape analogies. (a) Initial geometry showing critical overhang regions highlighted in lighter colors; (b) optimized geometry with surface normals relative to printing direction, maintaining structural topology while ensuring printability.

Algorithm 1 Optimization for Overhang Control

Input: V, F (mesh vertices and faces), λ (weight), *maxIter*, *tolerance*
Output: U (optimized vertices with controlled normals)

function PRECOMPUTATION(V, F, λ)
  Store mesh data: V, F, vertex normals N
  Compute vertex areas A and cotangent weights ▷ For energy terms
  Build vertex-face adjacency information
  for all vertex k **do**
    Collect adjacent faces and build edge list E_{fk}
    Store edge vectors and weights for deformation
  Construct sparse matrices Q_1, K_1 for global system
  $data.LHS \leftarrow Q_1/2$ ▷ For quadratic optimization
  return $data$
function FITROTATION($U, data$)
  Initialize rotation matrices and energy
  for all element k **do**
    Extract area a, edges E, weights W, vectors dV
    Compute deformation gradient dU
    Get current and desired normals n, n_{desire}
    $S \leftarrow dV \cdot W \cdot dU^T + \lambda \cdot a \cdot n \cdot n_{desire}^T$
    $R \leftarrow$ ORTHOGONALPROCRUSTES(S)
    Store rotation and update objective value
  return $RAll$, $objVal$, $data$
Initialize $U \leftarrow V$ and set print direction $v \leftarrow [1, 0, 0]$
Get boundary vertices b and conditions bc ▷ From mesh outline
$data \leftarrow$ PRECOMPUTATION(V, F, λ)
for $iter \leftarrow 1$ **to** $maxIter$ **do**
  ▷ Local Step: Compute Rotations
  $[RAll, objVal, data] \leftarrow$ FITROTATION($U, data$)
  ▷ Global Step: Update Vertices
  Reshape and solve system for new vertices U
  ▷ Update Normal Constraints
  $N \leftarrow$ COMPUTEVERTEXNORMALS(U, F)
  for all vertex i **do**
    $factor \leftarrow 2|\,|ang - \pi|/\pi - 0.5|$ ▷ Angle with print dir
    if $factor > 0.222$ **then**
      Project normal and set x-component to 0.364
  ▷ Check Convergence
  $relDV \leftarrow \max(\|U - U_{prev}\|)/\max(\|U - V\|)$
  if $relDV < tolerance$ **then break**
return U

challenges of 3DCP for spanning structures: establishing precise printing parameters, developing optimized toolpath strategies through computational slicing, preparing assembly components, and implementing post-tensioning techniques.

4.1. Printing setup and material

A two-component (2K) printing system, illustrated in Fig. 15, forms the core of the fabrication process. The setup consists of a material handling system (mixing pumps, storage, and air compressor), an ABB IRB 6640 robot equipped with a specialized print head for controlled deposition, and chemical metering units for accelerator addition. The print head's 2K mixing mechanism enables on-demand material property adjustment during printing. This system separates the concrete mixture into dry components (cement and aggregates) and liquid components (water and chemical additives), enabling precise rheological control throughout the printing sequence. The printing setup integrates multiple coordinated subsystems around an *ABB IRB 6640* industrial robot equipped with a specialized extruder from *Baumit*. The print head assembly features a 2K mixing mechanism where the separate components combine immediately before extrusion. This mixing approach allows for precise material property regulation, which is critical for maintaining consistent layer adhesion. Material delivery occurs via dedicated mortar mixing pumps for the primary concrete mixture and precision chemical metering pumps for the accelerator liquid. The entire system operates under hierarchical control protocols that ensure synchronized material flow throughout the printing process. Auxiliary systems support the primary printing operations, including an air compressor for material flow regulation and cleaning operations, operating on a 48-inch by 40-inch base platform.

The material system consists of *Baumit PrintCret 230* as the primary printing medium and *Baumit BauMinator AdSpeed 200* as the accelerator. *PrintCret 230*, supplied in 25 kg bags, is a specialized cement-based mortar with precisely defined technical properties according to EN 196-1 [64]: maximum grain size of 2.0 mm, dry bulk density of 1990 kg/m^3, bending strength of 10 MPa, and compressive strength of 55 MPa. Optimal printing consistency is achieved with a water content of 14–16%. The liquid accelerator, *BauMinator AdSpeed 200*, is specifically formulated for *PrintCret* mortar systems, with dosing precisely regulated through the printing system's metering pumps to ensure proper setting behavior.

The fabrication process employs carefully calibrated parameters to ensure consistent material deposition and layer bonding:

- **Layer height**: 10 mm nominal (range: 8–12 mm depending on material consistency and local geometry)

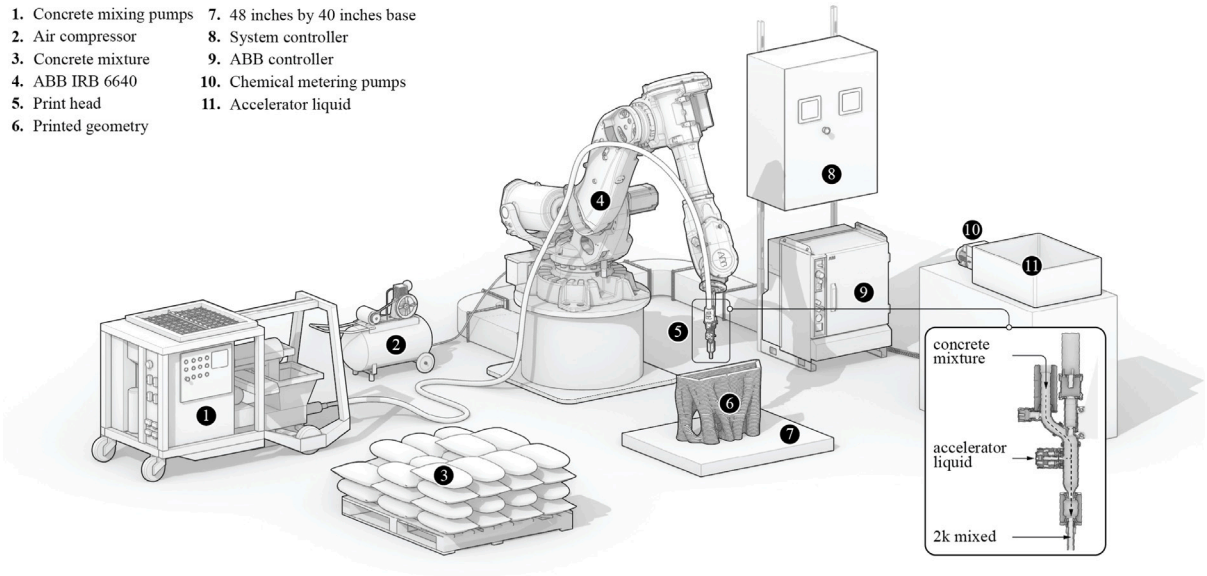


Fig. 15. 2K concrete printing system integrating material preparation, robotic deposition, and chemical acceleration components.

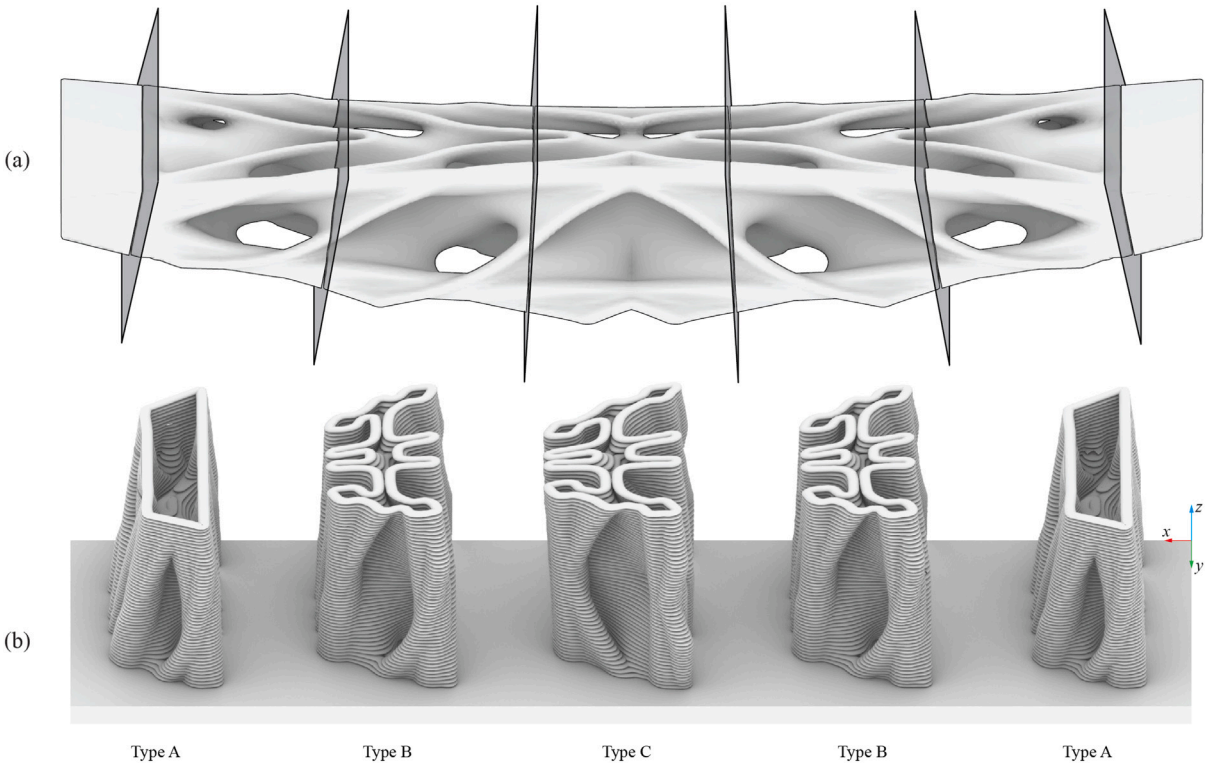


Fig. 16. Segmentation and printing simulation for the structure. (a) The division into five segments with three distinct typologies (Type A, B, and C); (b) representation of printed segments, showing deposition direction along the z-axis.

- **Layer width**: 30 mm (±2 mm variation based on extrusion pressure)
- **Print speed**: 80–120 mm/s (adjusted based on geometric complexity and overhang angles)
- **Accelerator dosage**: 2.5–3.5% (increased to 4% for overhanging sections requiring rapid strength gain)
- **Printing time**: 15–20 min per segment
- **Curing time**: 28 days (55 MPa compressive strength)

4.2. Slicing and toolpath generation

The implementation of the design necessitates a segmentation approach due to constraints imposed by the printing bed dimensions and manual handling requirements during assembly. This segmentation strategy, shown in Fig. 16(a), divides the structure into five distinct segments comprising three component types: two Type A — end sections, two Type B — intermediate sections, and one Type C — central section.

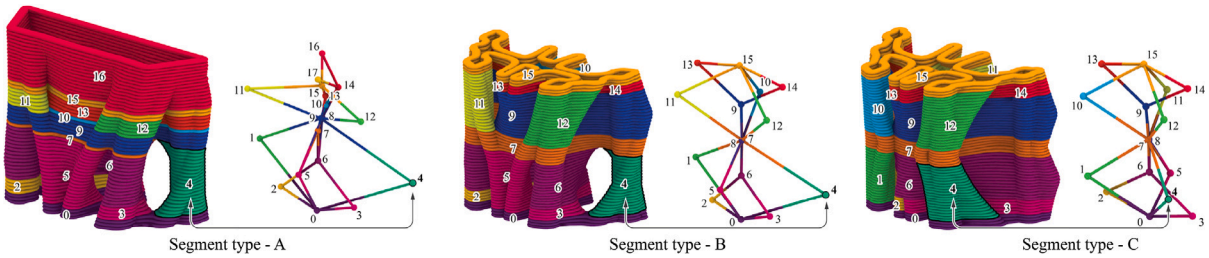


Fig. 17. Layer grouping strategy and topological relationships for each segment. Left to right: Type-A (end segment), Type-B (intermediate segment), and Type-C (central segment). Colors indicate distinct printing groups, with corresponding topological graphs showing connectivity between layer groups.

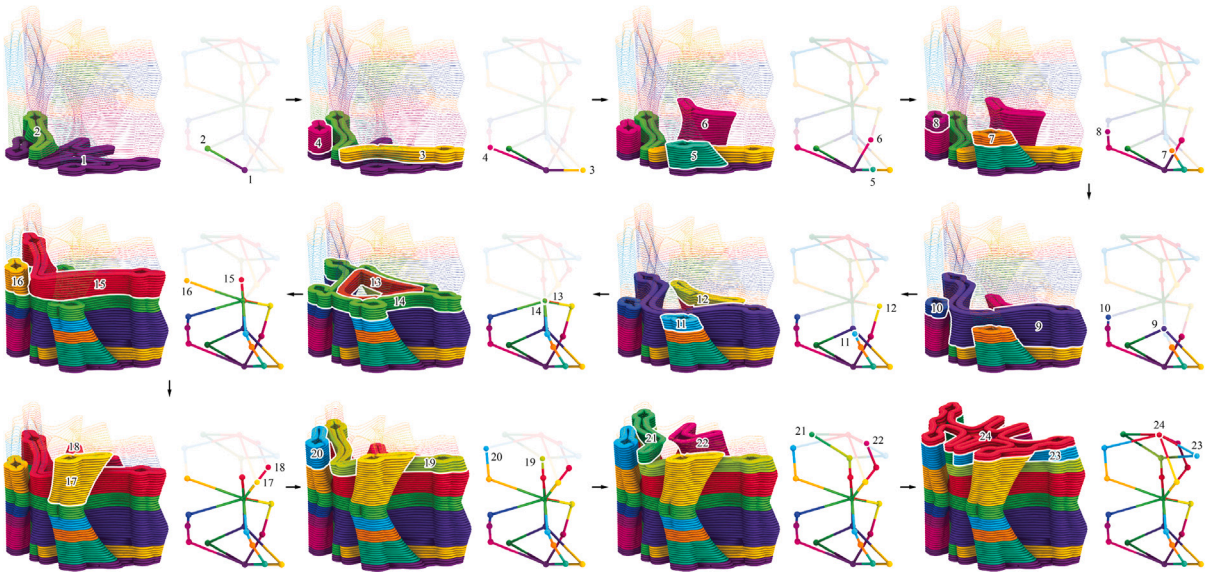


Fig. 18. Progressive toolpath generation for Type-C segment showing continuous printing sequence. Each step demonstrates layer group progression (1–24) with corresponding topological relationships. Colors indicate distinct printing groups, and graphs show optimized connectivity for nozzle clearance and path continuity.

The printing simulation is visualized in Fig. 16(b), where each segment is translated into a series of printing layers aligned with the z-axis printing direction. Our previous research [62] suggests that a porous geometry can be divided into vertically continuous patches. Their interdependency forms a directed acyclic graph whose topological sort can be a viable printing sequence. To avoid collision, we require that the highest point of the finished print and the lowest point of the unprinted toolpath differ by at most the nozzle height during the print. We also impose a constraint that the cylindrical nozzle should not collide with existing print, adding more directed edges to the graph [65]. Finally, a greedy algorithm subdivides and organizes the patches according to the two rules to form the printable continuous toolpath curves.

The process employs non-parallel slicing techniques to align layers perpendicular to compression forces. The fabrication process controls printing parameters to maintain consistent material deposition, with layer heights varying around a nominal 10 mm value and layer widths targeting 30 mm throughout the structure. Overhang management represents a critical challenge in the printing process. A maximum overhang angle of 35 degrees was established to ensure structural stability and proper layer adhesion. This constraint was implemented through comprehensive overhang optimization and continuously validated through analysis of surface normal vectors during the slicing process. The complex void distribution inherent in geometry requires the analysis of layer deposition sequences. Consequently, the printing sequence is organized into discrete groups of continuous layers, minimizing printing interruptions while maintaining geometric accuracy across void regions.

Fig. 17 demonstrates the organization of printing sequences for segment types A, B, and C. The printing strategy employs color-coded layer groups that optimize material deposition continuity based on geometric connectivity analysis. Each segment's topological graph maps the sequential relationships between layers, with numbered nodes indicating printing order and connecting lines showing direct paths between layer groups. Using the Type-C segment (Fig. 18) as an example, the greedy algorithm introduced in [62] optimizes the printing sequence while maintaining nozzle clearance requirements. The process progresses methodically from foundation layers (1–4) through intermediate sections (5–16) to completion layers (17–24). This structured approach ensures efficient transitions between layer groups while preventing nozzle collisions and maintaining consistent material flow. The integration of topological analysis with algorithmic path planning enables efficient material deposition while reducing printing interruptions. This methodology optimizes both the printing sequence and travel movements, ensuring manufacturing feasibility while preserving geometric accuracy.

The validation of the toolpath strategy is demonstrated through the fabrication of a Type-B segment. Fig. 19 documents the printing sequence, demonstrating continuous printing execution from foundation to completion, validating the toolpath strategy and material behavior during construction. The time-lapse sequence shows precise control over material deposition across varying overhang angles and surface transitions of isolated island parts. Fig. 20 demonstrates the final printed segments. The arrangement of printed components Fig. 20(a)

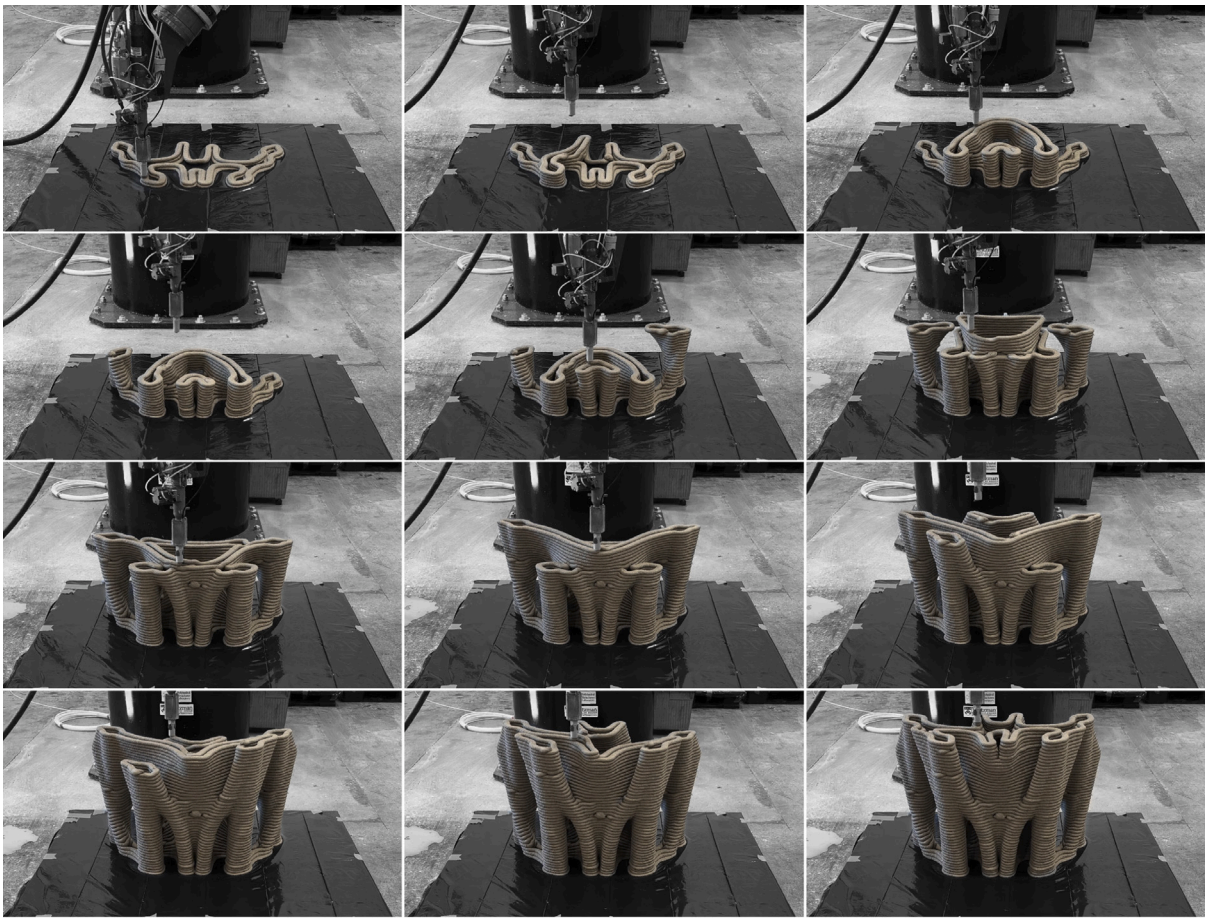


Fig. 19. Time-lapse documentation of Type-B segment fabrication showing progressive layer deposition of anticlastic geometry.

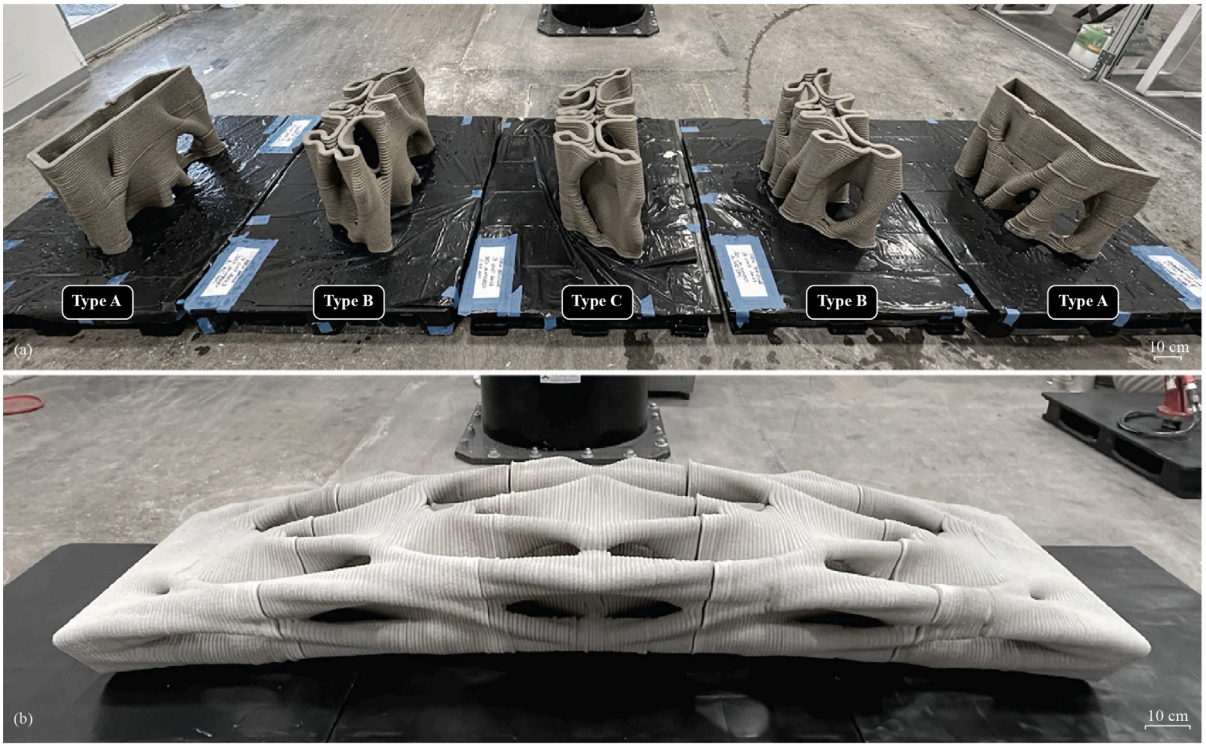


Fig. 20. (a) Three types of printed segments (Type A, B, and C); (b) a Three-meter span structure temporarily aligned, revealing continuous surface patterning and precise geometric alignment across segment interfaces prior to post-tensioning implementation.

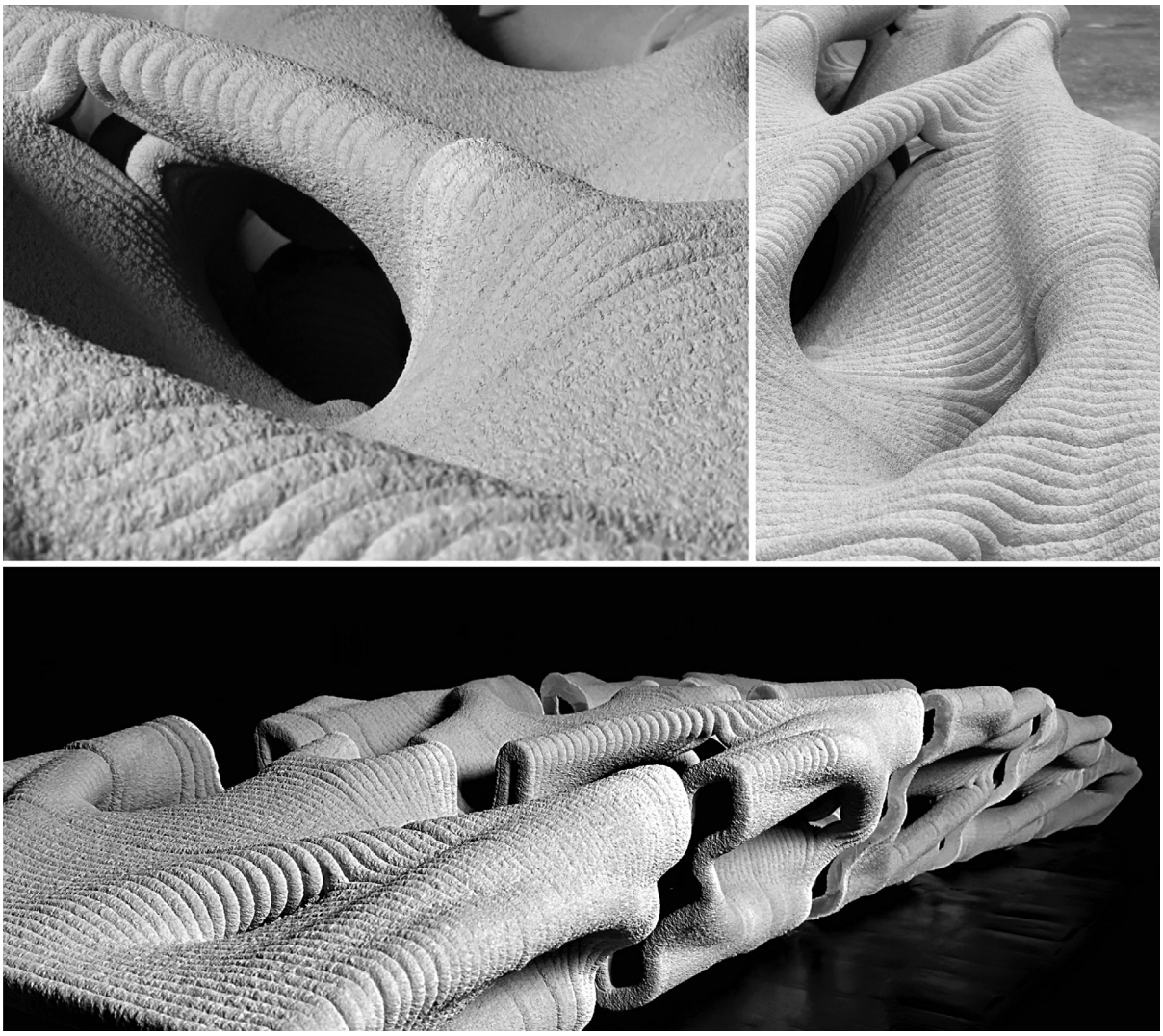


Fig. 21. Top: Close-up views revealing the relationship between printed segments and surface continuity. Bottom: Global perspective highlighting the integration of conduits within the geometry.

illustrates five segments with three structural types, where each shares identical layers with its adjacent segments. The geometric continuity is achieved through precise computational design, ensuring that the final layer of one segment perfectly matches the initial layer of the next, creating seamless transitions across the entire structure. The assembly validation process Fig. 20(b) verifies geometric alignment by positioning the segments upside-down, the flat upper surface of the structure provides a stable reference plane for assessing interface precision and surface continuity. The fabricated structure demonstrates geometric fidelity through its precisely articulated solid volumes and void spaces. Fig. 21 presents detailed close-up views revealing how the carefully calibrated layer deposition creates fluid surface transitions across complex curvatures. Voids, characteristic of anticlastic geometry, reduce material volume while maintaining structural integrity through the distribution of concrete material. The consistent layer patterning visible throughout the surface exemplifies how the printing methodology maintains uniform material thickness during critical geometric transitions between solid and void regions. The perspective view in Fig. 21 illustrates the global coherence of this solid–void relationship, demonstrating the topological integration of the pattern. Conduits for steel cables are incorporated in the geometry along the load lines associated with tension forces. These cables are post-tensioned to unify the segments. Further discussion of the construction is described in the next section.

4.3. Assembly and construction

Following the fabrication of individual segments through the printing process, the assembly phase introduces additional steps in completing the structure. The process involves the alignment of printed segments, the threading of steel cables through specified conduits, the integration of interface materials between segments, and the application of appropriate post-tensioning forces to achieve designated structural integration.

The assembly requires the integration of multiple components to achieve structural consolidation. Fig. 22 illustrates the comprehensive steps, where each element serves a specific structural and functional purpose. The post-tensioning jack Fig. 22(1) provides controlled force application essential for structural integration, while CNC milled form blocks Fig. 22(2) ensure proper alignment and support during the post-tensioning process. These components work in conjunction with the printed segments Fig. 22(3). The implementation of post-tensioning requires robust end anchorages to effectively transfer compression forces throughout the structure. In Fig. 22(3), the end-piece(Type A) was infilled with concrete and embedded reinforcement cages to ensure effective force transfer from the post-tensioning system while reducing local stress concentrations. The 0.5 inch diameter cables Fig. 22(5) run through the integrated conduit within the segments. To

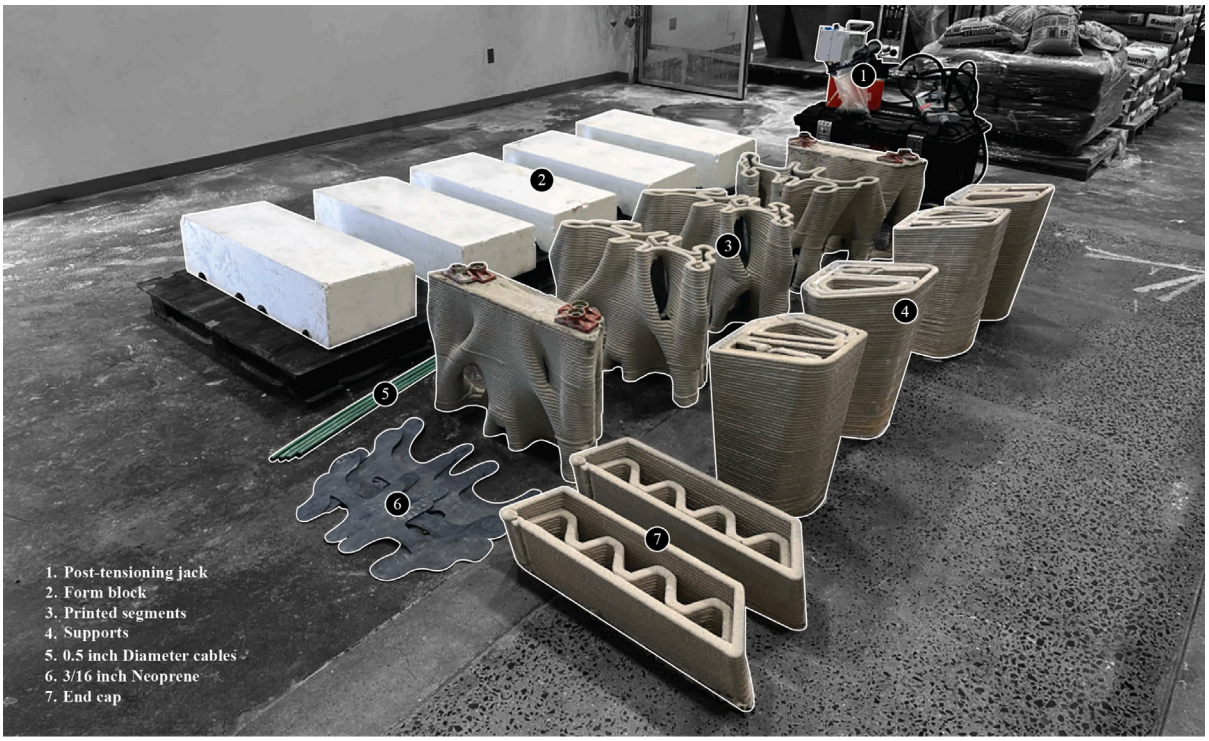


Fig. 22. Components for assembly of a three meter spanning structure:(1) post-tensioning jack for force application, (2) foam blocks for alignment, (3) 3D concrete printed segments, (4) support pedestals, (5) 12.7 mm diameter post-tensioning cables, (6) 4.76 mm neoprene, and (7) end caps.



Fig. 23. Assembled 3D concrete printed, post-tensioned anticlastic concrete structure realized through the application of the comprehensive workflow that develops upon and integrates computational design with digital and physical fabrication as well as construction.

protect the concrete interfaces and ensure uniform load distribution, 3/16 inch neoprene pad (Fig. 22(6)) are strategically cut and placed between segments. This arrangement of components enables the transition from individually printed elements to a unified structural system. The assembly is completed with end caps and supports shown in Fig. 23.

The validation through prototype fabrication at multiple scales reveals considerations for translating computationally designed geometries into physical structures. While small-scale PLA prototypes (Fig. 13) demonstrate precise geometric realization with smooth surface finishes characteristic of thermoplastic printing, the three-meter concrete structure (Figs. 21, 23) exhibits surface qualities inherent to

cementitious material extrusion at structural scales. The increase in layer height (0.2 mm vs. 10 mm) produces visible layer articulation in the concrete structure, while the time-dependent rheology of fresh concrete creates subtle surface smoothing absent in rapidly-solidifying thermoplastics. Despite the layer height difference, concrete printing achieves significantly shorter total fabrication time: each segment requires approximately 2–3 h compared to 15–20 h for equivalent PLA prototypes at scale. These material and scale dependent surface variations, both prototypes validate the workflow's effectiveness across scales. The visible layer stratification in the concrete structure records the robotic toolpath and material deposition sequence, establishing a direct correspondence between digital fabrication logic and physical surface articulation.

5. Conclusions and future work

This paper presented a comprehensive workflow that integrates computational design and digital fabrication for complex spanning structures through 3DCP. The overall workflow demonstrated three key contributions: first, a systematic approach to form-finding that effectively combines tension and compression elements; second, the successful implementation of embedding periodic anticlastic geometry; and third, a validated fabrication strategy that enables discrete printed elements to function as a cohesive structural system, as demonstrated through a three-meter beam prototype.

The workflow integrated structural form-finding, geometric materialization, and fabrication optimization into a unified design approach. The PGS methodology provided precise control over force distribution, enabling effective allocation of materials by identifying compression and tension regions to respectively place concrete and steel. The embedding of anticlastic geometries within PGS-informed cells created structural forms with optimal material distribution possible through 3DCP. Practical fabrication constraints of 3DCP are addressed through customized algorithms that minimize overhangs and generate continuous toolpaths for printing complex geometries.

The practical implementation of the workflow was validated through the successful design and construction of a three-meter spanning beam prototype. This physical demonstration confirmed the feasibility of the approach, particularly in translating computationally derived geometries into discrete printable elements that function as a structural system when assembled. The prototype successfully demonstrated how combining post-tensioning techniques with discrete 3DCP modules allows structures to exceed the build volume of 3D concrete printers, enabling the construction of longer spans by printing smaller components that are then assembled and tensioned together into a unified structure.

The feasibility of this integrated approach has been demonstrated through the experimental validation of a 4.5-meter funicular beam prototype designed using the same computational framework [66]. Four-point bending tests confirmed the structural capacity (32.1 kN ultimate load) and revealed the complex behavior of modular post-tensioned systems with embedded anticlastic geometry. The prototype's performance validated that the funicular geometry effectively directed compression forces through concrete segments while post-tensioning cables carried tensile loads, and the modular assembly strategy enabled both structural performance and end-of-life recyclability through complete material separation.

The computational workflow was applied to a 10-meter beam to further evaluate material efficiency at architectural scales. Table 1 compares this form-found beam with a conventional reinforced concrete beam designed for equivalent 10-meter span and load capacity.

The form-found beam employs a wider, shallower section (1.88 m × 0.76 m, span-to-depth 13.26) with 74.0% void ratio, while conventional design uses solid cross-section (0.45 m × 1.0 m, span-to-depth 10.0). The form-found beam achieves 34.1% weight reduction (7526.5 kg vs. 11,436 kg). Normalized by width, material efficiency reaches 84.3% reduction (4001 kg/m vs. 25,413 kg/m).

This efficiency results from integrated strategies: strategic void distribution utilizes 3.75 m^3 concrete within 14.40 m^3 envelope; PrintCret 230 density (1990 kg/m^3) reduces self-weight versus standard concrete (2400 kg/m^3); and PGS-guided material placement enables eight post-tensioning cables (64 kg) to replace 636 kg distributed reinforcement—89.9% steel reduction. The reduced mid-span depth (0.76 m vs. 1.0 m) decreases floor-to-floor height requirements by 24 cm per floor. Beyond material savings, 3DCP eliminates formwork, achieves 20-minute fabrication per segment versus 2–3 day formwork assembly, and enables potential material separation for recycling.

Future work will focus on scaling this methodology to larger structural applications, exploring alternative material systems and their integration with anticlastic geometry, and optimizing post-tensioning strategies for enhanced structural performance. Additional investigations will examine adaptive design variations for different architectural applications and develop automated assembly processes for more complex geometrical configurations. These developments aim to expand the implementation of computationally designed concrete structures while maintaining the crucial balance between geometric complexity, structural efficiency, and constructability that this work has established.

CRediT authorship contribution statement

Hua Chai: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Mathias Bernhard:** Visualization, Software, Methodology, Formal analysis. **Yefan Zhi:** Writing – original draft, Visualization, Software, Methodology. **Maximilian E. Ororbia:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Data curation, Conceptualization. **Masoud Akbarzadeh:** Writing – review & editing, Validation, Supervision, Software, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Masoud Akbarzadeh reports financial support was provided by National Science Foundation. Masoud Akbarzadeh reports financial support was provided by Advanced Research Projects Agency-Energy. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

Table 1
Comparison between form-found beam with embedded anticlastic geometry and conventional reinforced concrete beam for 10-meter span application.

Parameter	Form-found beam	Conventional RC beam
Geometric Properties		
Span length (m)	10.08	10.0
Width (m)	1.88 (end)	0.45
Depth (m)	0.76 (mid)	1
Span-to-depth ratio	13.26	10.0
Envelope volume (m^3)	14.4	4.5
Material Quantities		
Solid concrete volume (m^3)	3.75	4.5
Concrete density (kg/m^3)	1990	2400
Concrete weight (kg)	7462.5	10,800
Steel reinforcement (kg)	64 (8 cables)	636 (distributed rebar)
Total weight (kg)	7526.5	11,436
Material Efficiency		
Void ratio (%)	74	0
Steel ratio (%)	0.85	5.6
Weight per meter width (kg/m)	4001	25,413
Performance Comparison		
Total weight reduction	34.2%	Baseline
Steel reduction	89.9%	Baseline
Concrete reduction	30.9%	Baseline
Construction Characteristics		
Formwork required	None	Full formwork
Assembly approach	Post-tensioned segments	Cast-in-place

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